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A Remarkable Representation of the 3 + 2 de Sitter Group

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Among the infinitesimal operators of the 3 + 2 de Sitter group, there are four independent cyclic ones, one of which is separate from the other three. A representation is obtained for which this one has integral eigenvalues while the other three have half-odd eigenvalues, or vice versa. The representation is of a specially simple kind, with the wavefunctions involving only two variables.

INTRODUCTION

We consider the group of rotations of five real variables x_1, x_2, x_3, x_4, x_5 which leave the quadratic form

$$x_1^2 + x_2^2 + x_3^2 - x_4^2 - x_5^2 \tag{1}$$

invariant. The infinitesimal operators of the group are $m_{ab} = -m_{ba}(a, b = 1, 2, 3, 4, 5)$. There are ten independent ones. They satisfy the following commutation relations, in the notation $[\xi, \eta] = \xi\eta - \eta\xi$, with a, b, c, d all different:

$$[m_{ab}, m_{cd}] = 0. \tag{2}$$

With a, b, c, all different and no summation over a,

$$\begin{aligned} [m_{ab}, m_{ac}] &= m_{bc} \quad \text{for} \quad a = 4, 5, \\ &= -m_{bc} \quad \text{for} \quad a = 1, 2, 3. \end{aligned} \tag{3}$$

We shall consider only unitary representations, so that im_{ab} has real eigenvalues, for all a, b.

Some of the basic rotations m_{ab} are cyclic and some are hyperbolic. The cyclic ones are m_{12}, m_{23}, m_{31}, m_{45}, and the hyperbolic ones are m_{14}, m_{24}, m_{34}, m_{15}, m_{25}, m_{35}. For m_{ab} hyperbolic, im_{ab} has a continuous range of eigenvalues and for m_{ab} cyclic, im_{ab} has discrete eigenvalues, which may be either integers or half-odd integers.

If we set up wave equations in the 3 + 2 de Sitter space,

$$x_1^2 + x_2^2 + x_3^2 - x_4^2 - x_5^2 = -R^2, \tag{4}$$

with various tensor or spinor kinds of wavefunctions, we get various representations of the de Sitter group, but they are such that, for all the cyclic operators, m_{ab}, im_{ab} has integral eigenvalues, or else for all of them it has half-odd integral eigenvalues. For these straightforward representations, a mixing of integral and half-odd integral eigenvalues does not occur.

The present paper is concerned with a more primitive representation, for which im_{12}, im_{23}, im_{31} have half-odd integral eigenvalues while im_{45} has integral eigenvalues. There is nothing inconsistent in such a mixing, because the m_{45} rotation is completely detached from the m_{12}, m_{23}, m_{31} rotations. In fact, if one goes over to the covering group of the 3 + 2 de Sitter group, the detachment of the m_{45} rotation allows im_{45} to have any real eigenvalues, independently of what eigenvalues the other rotations have. A general theory of the representations of this covering group has been given by Ehrman.[1]

THE γ MATRICES

We consider 4×4 matrices whose elements are all real. There are 16 independent ones. We choose them in a certain way and call them γ matrices. One of them is the unit matrix γ_0. The other 15 are chosen to be all symmetric or skew matrices, with the square of each symmetric one equal to γ_0 and the square of each skew one equal to $-\gamma_0$, and with the product of any two equal to $\pm$ a third one. These 15 may be written $\gamma_{AB} = -\gamma_{BA}$, where A and B are two different suffixes going from 1 to 6.

The rules for multiplying the γ_{AB} are as follows: We use the notation $\gamma_{AB}\gamma_{CD} = \gamma_{ABCD}$, and so on for products of more than two factors. Thus any product appears as a γ with an even number of suffixes. There are two general rules: (i) Any two different suffixes may be interchanged, if one brings in the factor -1. (ii) A suffix A occurring in two consecutive positions may be suppressed but one must then bring in the factor -1 for $A = 4$, 5 or 6. Thus for example, with $a, b = 1, \cdots 5$,

$$\gamma_{a6}\gamma_{b6} = \gamma_{a6b6} = -\gamma_{ab66} = \gamma_{ab}. \tag{5}$$

The suppression of all the suffixes yields γ_0.

As a consequence of these rules, we find $\gamma_{AB}^2 = \gamma_0$ if one of the suffixes A, B is in the set 1, 2, 3, and

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[1] J. B. Ehrman, Proc. Cambridge Phil. Soc. 53, 290 (1957).

the other in the set 4, 5, 6, so γ_{AB} is then symmetric. Otherwise $\gamma_{AB}^2 = -\gamma_0$ and γ_{AB} is skew. We find also that γ_{AB} and γ_{CD} commute if A, B, C, D are all different and anticommute if one of the suffixes A, B is the same as one of the suffixes C, D. Thus a set of five γ_{AB}'s with one suffix in common, all anticommute. Further we find that

$$\gamma_{123456}^2 = \gamma_0,$$

so that

$$\gamma_{123456} = \pm\gamma_0.$$

We arrange that

$$\gamma_{123456} = \gamma_0. \tag{6}$$

This equation enables us to reduce γ_{ABCD} to a two-suffix γ when A, B, C, D are all different. For example, multiplying (6) by $-\gamma_{56}$, we get

$$\gamma_{1234} = -\gamma_{56}.$$

The reduction of γ_{ABCD} to a two-suffix γ when two of the suffixes A, B, C, D are the same is given directly by the rules (i) and (ii), as is illustrated by the example (5).

The result of interchanging rows and columns in any matrix α will be written $\alpha^\dagger$. We have $\gamma_{AB}^\dagger = \pm\gamma_{AB}$, the + sign occurring when $\gamma_{AB}^2 = \gamma_0$ and the − sign when $\gamma_{AB}^2 = -\gamma_0$.

Each γ_{AB} has the diagonal sum zero;

$$\langle\gamma_{AB}\rangle = 0. \tag{7}$$

To prove this, let σ be any 4×4 matrix with a reciprocal. Then

$$\langle\gamma_{AB}\rangle = \langle\gamma_{AB}\sigma\sigma^{-1}\rangle = \langle\sigma^{-1}\gamma_{AB}\sigma\rangle.$$

By taking σ to be one of the γ matrices that anticommutes with γ_{AB}, we get

$$\langle\gamma_{AB}\rangle = -\langle\gamma_{AB}\rangle.$$

A REPRESENTATION OF THE CONFORMAL GROUP

Introduce four variables u_1, u_2, u_3, u_4 to correspond to the four rows of the γ matrices. They may be written as a column matrix and we then denote them by the symbol u. Alternatively, they may be written as a row matrix and are then denoted by $u^\dagger$.

Let v_1, v_2, v_3, v_4 be a second such set of four variables. With α any 4×4 matrix, we may form, by matrix multiplication, $v^\dagger\alpha u$, which is a 1×1 matrix or a number. We have evidently

$$v^\dagger\alpha u = u^\dagger\alpha^\dagger v. \tag{8}$$

Introduce the four differential operators $\partial_n = \partial/\partial u_n (n = 1, 2, 3, 4)$. They satisfy the commutation relations

$$[\partial_n, u_{n'}] = \delta_{n,n'}. \tag{9}$$

We may write them as a column matrix ∂ or as a row matrix $\partial^\dagger$. If we substitute ∂ for v in (8), the equation is no longer valid, on account of the lack of commutation of the ∂'s and u's. One easily finds from (9),

$$\partial^\dagger\alpha u = u^\dagger\alpha^\dagger\partial + \langle\alpha\rangle. \tag{10}$$

From (8) and (9) we find also, for α and β any two 4×4 matrices,

$$[u^\dagger\alpha\partial, u^\dagger\beta\partial] = u^\dagger[\alpha, \beta]\partial. \tag{11}$$

For each matrix γ_{AB}, define the differential operator

$$\chi_{AB} = \tfrac{1}{2}u^\dagger\gamma_{AB}\partial. \tag{12}$$

From (11) we get

$$[\chi_{AB}, \chi_{CD}] = \tfrac{1}{4}u^\dagger(\gamma_{ABCD} - \gamma_{CDAB})\partial.$$

It follows that, for A, B, C, D all different,

$$[\chi_{AB}, \chi_{CD}] = 0, \tag{13}$$

and for A, B, C all different, with no summation over A,

$$\begin{aligned}[\chi_{AB}, \chi_{AC}] &= \chi_{BC} \quad \text{for} \quad A = 4, 5, 6,\\ &= -\chi_{BC} \quad \text{for} \quad A = 1, 2, 3.\end{aligned} \tag{14}$$

The commutation relations (13), (14) are like the commutation relations (2), (3), with the difference that the suffixes can take on six values instead of five. They show that the χ_{AB} may be looked upon as infinitesimal operators of the group of rotations of six real variables $x_1 \cdots x_6$ which leave

$$x_1^2 + x_2^2 + x_3^2 - x_4^2 - x_5^2 - x_6^2 \tag{15}$$

invariant. This is the conformal group of four-dimensional space. It contains the 3 + 2 de Sitter group as a subgroup.

With the u's restricted to be real variables, the χ_{AB} provide a representation of the conformal group for which the wavefunctions are functions $\psi(u_1, u_2, u_3, u_4)$ of the four real u's. The representation is unitary, because the adjoint of χ_{AB} is

$$-\tfrac{1}{2}\partial^\dagger\gamma_{AB}^\dagger u = -\chi_{AB},$$

from (10) and (7). If we choose ψ so that $\int |\psi|^2\, d^4u$ converges, the integral is left invariant by the application of any infinitesimal rotation to ψ.

Corresponding to (12), we define

$$\chi_0 = \tfrac{1}{2}u^\dagger\gamma_0\partial = \tfrac{1}{2}u^\dagger\partial.$$

We see from (11) that χ_0 commutes with every χ_{AB}. Thus we may put χ_0 equal to a number, a, say. The condition that $\chi_0 = a$ means that the wavefunction ψ is homogeneous in the u's of degree $2a$. Let us introduce polar variables ρ, θ_1, θ_2, θ_3 in the u space, with $\rho = (u^\dagger u)^{\frac{1}{2}}$, and θ_1, θ_2, θ_3, three independent functions of the ratios of the u's. Then ψ is of the form

$$\psi = \rho^{2a}\psi_1(\theta_1, \theta_2, \theta_3). \tag{16}$$

To secure the unitary character of the representation, we require

$$i(u\partial^\dagger + \partial^\dagger u) = \text{real number}.$$

Now,

$$u^\dagger\partial + \partial^\dagger u = 2u^\dagger\partial + 4 = 4(\chi_0 + 1).$$

Thus,

$$\chi_0 = -1 + ib, \tag{17}$$

where b is a real number. Equation (16) becomes

$$\psi = \rho^{-2+2ib}\psi_1(\theta). \tag{18}$$

The wavefunction ψ is now not square integrable, being an eigenfunction of an operator χ_0 with a continuous range of eigenvalues. We have

$$\int |\psi|^2\, d^4u = \int |\psi|^2\, \rho^3\, d\Omega\, d\rho,$$

where $d\Omega$ is an element of 3-dimensional solid angle in the u space and is some multiple of $d\theta_1\, d\theta_2\, d\theta_3$. Substituting the form (18) for ψ, we get

$$\int_{-\infty}^{\infty} |\psi|^2\, d^4u = \int_0^\infty \rho^{-1}\, d\rho \int |\psi_1|^2\, d\Omega. \tag{19}$$

We may now drop the variable ρ from the representation, which involves working with the wave function $\psi_1(\theta)$ and dropping the infinite factor $\int \rho^{-1}\, d\rho$ from the formula (19) for the squared length of a wavefunction.

Going back to general wavefunctions ψ that are not eigenfunctions of χ_0, let us consider wavefunctions of the form

$$\psi = P(u_1, u_2, u_3, u_4)f(\rho), \tag{20}$$

where P is a power series in the u's and $f(\rho)$ is chosen so that ψ is square integrable. One finds readily

$$[\chi_{AB}, \rho^2] = u^\dagger\gamma_{AB}u. \tag{21}$$

If the rotation χ_{AB} is cyclic, γ_{AB} is skew, and so χ_{AB} commutes with ρ. Thus a cyclic rotation applied to the wavefunction (20) affects only the power series P and leaves the factor $f(\rho)$ invariant.

Take P to be linear in the u's and suppose it is an eigenfunction of a particular cyclic χ_{AB}. The eigenvalue is then an eigenvalue of $\frac{1}{2}\gamma_{AB}$, and is thus $\pm\frac{1}{2}i$. One can readily infer that, if P is homogeneous of the nth degree in the u's and is an eigenfunction of a particular cyclic χ_{AB}, the eigenvalue of $i\chi_{AB}$ is an integer or half-odd integer according to whether n is even or odd.

The parity of the power series P is invariant under all the rotations, hyperbolic as well as cyclic. This parity determines the integral or half-odd integral character of the eigenvalues of each of the cyclic rotations. It follows that in the present representation there is no mixing of integral and half-odd integral eigenvalues, such as occurs in the remarkable representation we are seeking.

SOME PROPERTIES OF THE OPERATORS χ

Lemma. For any symmetrical 4×4 matrix S,

$$\gamma_{12}S\gamma_{12} + \gamma_{23}S\gamma_{23} + \gamma_{31}S\gamma_{31} = S - \langle S\rangle\gamma_0, \tag{22}$$

and similarly,

$$\gamma_{45}S\gamma_{45} + \gamma_{56}S\gamma_{56} + \gamma_{64}S\gamma_{64} = S - \langle S\rangle\gamma_0. \tag{23}$$

To prove (22), we note that any symmetric γ_{AB} has one of its suffixes equal to 1, 2, or 3 and the other equal to 4, 5, or 6 and therefore it commutes with one of the three quantities γ_{12}, γ_{23}, γ_{31} and anticommutes with the other two. Thus if such a γ_{AB} is substituted for S in (22), two of the terms on the left become S and the third one $-S$, and the $\langle S\rangle$ on the right is zero. Also (22) holds with $S = \gamma_0$, so it holds for any symmetric S. The proof of (23) is similar.

The lemma enables one to deduce a number of quandratic relations between the χ's. From (10),

$$2\chi_{AB} = u^\dagger\gamma_{AB}\partial = \partial^\dagger\gamma_{AB}^\dagger u.$$

Hence

$$-4(\chi_{12}^2 + \chi_{23}^2 + \chi_{31}^2) = \partial^\dagger\gamma_{12}uu^\dagger\gamma_{12}\partial + \partial^\dagger\gamma_{23}uu^\dagger\gamma_{23}\partial + \partial^\dagger\gamma_{31}uu^\dagger\gamma_{31}\partial.$$

The multiplication is associative, so we can pick out the $uu^\dagger$ in the middle of each term on the right and consider it as a 4×4 matrix. It is symmetric, so we can take it to be the S of (22) and get

$$-4(\chi_{12}^2 + \chi_{23}^2 + \chi_{31}^2) = \partial^\dagger uu^\dagger\partial - \partial^\dagger\langle uu^\dagger\rangle\partial = (u^\dagger\partial + 4)u^\dagger\partial - u^\dagger u\partial^\dagger\partial - 2u^\dagger\partial.$$

Thus,

$$\chi_{12}^2 + \chi_{23}^2 + \chi_{31}^2 = -\chi_0(\chi_0 + 1) + \tfrac{1}{4}u^\dagger u\partial^\dagger\partial. \tag{24}$$

Similarly,

$$\chi_{45}^2 + \chi_{56}^2 + \chi_{64}^2 = -\chi_0(\chi_0 + 1) + \tfrac{1}{4}u^\dagger u\partial^\dagger\partial. \tag{25}$$

Hence,

$$\chi_{12}^2 + \chi_{23}^2 + \chi_{31}^2 = \chi_{45}^2 + \chi_{56}^2 + \chi_{64}^2. \tag{26}$$

A more general result may be obtained as follows: Let γ_P and γ_Q be any two γ matrices, i.e., each of them is either γ_0 or a γ_{AB}. Putting $S = uu^\dagger$ in (22), and multiplying by $\partial^\dagger\gamma_P^\dagger$ on the left and $\gamma_Q\partial$ on the right, we get

$$\begin{aligned}\partial^\dagger\gamma_P^\dagger\gamma_{12}uu^\dagger\gamma_{12}\gamma_Q\partial &+ \partial^\dagger\gamma_P^\dagger\gamma_{23}uu^\dagger\gamma_{23}\gamma_Q\partial \\ &+ \partial^\dagger\gamma_P^\dagger\gamma_{31}uu^\dagger\gamma_{31}\gamma_Q\partial - \partial^\dagger\gamma_P^\dagger uu^\dagger\gamma_Q\partial \\ &= -\partial^\dagger\gamma_P^\dagger\langle uu^\dagger\rangle\gamma_Q\partial \\ &= -u^\dagger u\partial^\dagger\gamma_{P^\dagger Q}\partial - 2u^\dagger\gamma_{P^\dagger Q}\partial,\end{aligned}$$

where $\gamma_{P^\dagger Q}$ denotes $\gamma_{P^\dagger}\gamma_Q$. With the help of (10) and (12), this becomes

$$\begin{aligned}&\{\chi_{12P} + \tfrac{1}{2}\langle\gamma_{12P}\rangle\}\chi_{12Q} + \{\chi_{23P} + \tfrac{1}{2}\langle\gamma_{23P}\rangle\}\chi_{23Q} \\ &\quad + \{\chi_{31P} + \tfrac{1}{2}\langle\gamma_{31P}\rangle\}\chi_{31Q} + \{\chi_P + \tfrac{1}{2}\langle\gamma_P\rangle\}\chi_Q \\ &= \tfrac{1}{4}u^\dagger u\partial^\dagger\gamma_{P^\dagger Q}\partial + \chi_{P^\dagger Q}.\end{aligned} \tag{27}$$

Now $\langle\gamma_{12P}\rangle$ vanishes unless $\gamma_{12P} = \pm\gamma_0$, in which case $\gamma_{12} = \mp\gamma_P = \pm\gamma_{P^\dagger}$. Thus,

$$\langle\gamma_{12P}\rangle\chi_{12Q} = |\langle\gamma_{12P}\rangle|\,\chi_{P^\dagger Q}.$$

Treating the other $\langle\gamma\rangle$ terms in (27) in the same way, we get finally

$$\begin{aligned}&\chi_{12P}\chi_{12Q} + \chi_{23P}\chi_{23Q} + \chi_{31P}\chi_{31Q} + \chi_P\chi_Q \\ &\quad = \{1 - \tfrac{1}{2}\,|\langle\gamma_{12P} + \gamma_{23P} + \gamma_{31P} + \gamma_P\rangle|\}\chi_{P^\dagger Q} \\ &\qquad + \tfrac{1}{4}u^\dagger u\partial^\dagger\gamma_{P^\dagger Q}\partial.\end{aligned} \tag{28}$$

Similarly, it may be shown that

$$\begin{aligned}&\chi_{45P}\chi_{45Q} + \chi_{56P}\chi_{56Q} + \chi_{64P}\chi_{64Q} + \chi_P\chi_Q \\ &\quad = \{1 - \tfrac{1}{2}\,|\langle\gamma_{45P} + \gamma_{56P} + \gamma_{64P} + \gamma_P\rangle|\}\chi_{P^\dagger Q} \\ &\qquad + \tfrac{1}{4}u^\dagger u\partial^\dagger\gamma_{P^\dagger Q}\partial.\end{aligned} \tag{29}$$

It should be noted that the last term in (28) or (29) vanishes if $\gamma_{PQ}^\dagger$ or γ_{PQ} is skew.

There are many applications of the general formulas (28) and (29). With $\gamma_P = \gamma_Q = \gamma_0$, we get back to (24) and (25). With $\gamma_P = \gamma_0$, $\gamma_Q = \gamma_{56}$ in (28), we get

$$\chi_{12}\chi_{34} + \chi_{23}\chi_{14} + \chi_{31}\chi_{24} = -(1 + \chi_0)\chi_{56}. \tag{30}$$

With $\gamma_P = \gamma_{12}$, $\gamma_Q = \gamma_{45}$ in (28), we get

$$\begin{aligned}-\chi_0\chi_{36} + \chi_{31}\chi_{16} + \chi_{32}\chi_{26} + \chi_{12}\chi_{45} \\ = \chi_{36} - \tfrac{1}{4}u^\dagger u\partial^\dagger\gamma_{36}\partial.\end{aligned} \tag{31}$$

With $\gamma_P = \gamma_{31}$, $\gamma_Q = \gamma_{16}$ in (29), we get

$$\begin{aligned}\chi_{26}\chi_{32} + \chi_{24}\chi_{51} + \chi_{25}\chi_{14} + \chi_{31}\chi_{16} \\ = -\chi_{36} - \tfrac{1}{4}u^\dagger u\partial^\dagger\gamma_{36}\partial.\end{aligned} \tag{32}$$

Subtracting (32) from (31), we get (taking into account the noncommutativity of χ_{26} and χ_{32}),

$$\chi_{12}\chi_{45} + \chi_{24}\chi_{15} + \chi_{41}\chi_{25} = (1 + \chi_0)\chi_{36}. \tag{33}$$

Equations (30) and (33) and similar equations lead to the general formula, for A, B, C, D all different:

$$\chi_{AB}\chi_{CD} + \chi_{BC}\chi_{AD} + \chi_{CA}\chi_{BD} = (1 + \chi_0)\chi_{ABCD}. \tag{34}$$

With $\gamma_P = \gamma_Q = \gamma_{45}$ in (28), we get

$$\chi_{36}^2 + \chi_{16}^2 + \chi_{26}^2 + \chi_{45}^2 = \chi_0 + \tfrac{1}{4}u^\dagger u\partial^\dagger\partial. \tag{35}$$

Subtracting (25) from (35), we get

$$\chi_{16}^2 + \chi_{26}^2 + \chi_{36}^2 - \chi_{46}^2 - \chi_{56}^2 = \chi_0(\chi_0 + 2). \tag{36}$$

More generally, we have for each value of B,

$$\sum_{A\neq B}\chi_{AB}\chi^{AB} = -\chi_0(\chi_0 + 2), \tag{37}$$

where the raising of a suffix involves a minus sign if the suffix is 4, 5, or 6. Summing for all B, we get

$$\sum_A\sum_{B<A}\chi_{AB}\chi^{AB} = -3\chi_0(\chi_0 + 2), \tag{38}$$

THE REMARKABLE REPRESENTATION

With $\mu, \nu = 1, 2, 3, 4$, define

$$\begin{aligned}m_{\mu\nu} &= \chi_{\mu\nu}, \\ m_{\mu 5} &= C(\chi_{\mu 5} - i\chi_{\mu 6}),\end{aligned} \tag{39}$$

where C is the operator which converts numbers into their conjugate complexes. It has the algebraic properties

$$C^2 = 1, \qquad Ci = -iC, \tag{40}$$

and C commutes with all the ∂'s, u's and γ's. Thus it commutes with all the χ's.

We must check whether the m_{ab} defined by (39) satisfy the commutation relations (2), (3). It is evident from (14) that all these relations are satisfied except the ones for $[m_{\mu 5}, m_{\nu 5}]$ with $\mu \neq \nu$. We have

$$\begin{aligned}[m_{\mu 5}, m_{\nu 5}] &= C(\chi_{\mu 5} - i\chi_{\mu 6})C(\chi_{\nu 5} - i\chi_{\nu 6}) \\ &\quad - C(\chi_{\nu 5} - i\chi_{\nu 6})C(\chi_{\mu 5} - i\chi_{\mu 6}) \\ &= (\chi_{\mu 5} + i\chi_{\mu 6})(\chi_{\nu 5} - i\chi_{\nu 6}) \\ &\quad - (\chi_{\nu 5} + i\chi_{\nu 6})(\chi_{\mu 5} - i\chi_{\mu 6}) \\ &= [\chi_{\mu 5}, \chi_{\nu 5}] + [\chi_{\mu 6}, \chi_{\nu 6}] \\ &\quad + 2i(\chi_{\mu 6}\chi_{\nu 5} - \chi_{\mu 5}\chi_{\nu 6}) \\ &= 2\chi_{\mu\nu} + 2i\{-\chi_{\mu\nu}\chi_{56} + (1 + \chi_0)\chi_{\mu\nu 56}\}\end{aligned}$$

from (34), with $A, B, C, D = \mu, \nu, 5, 6$. Thus,

$$[m_{\mu 5}, m_{\nu 5}] = \chi_{\mu\nu} - 2i\chi_{\mu\nu}(\chi_{56} + \tfrac{1}{2}i) + 2i\chi_{\mu\nu 56}(1 + \chi_0). \quad (41)$$

Consider wavefunctions ψ that satisfy the two supplementary conditions

$$(\chi_0 + 1)\psi = 0, \quad (42)$$

$$(\chi_{56} + \tfrac{1}{2}i)\psi = 0. \quad (43)$$

The conditions are consistent since χ_0 and χ_{56} commute. The operators $\chi_0 + 1$ and $\chi_{56} + \frac{1}{2}i$ occurring in these conditions commute with all the m_{ab}. This is obvious for the operator $\chi_0 + 1$. It is also obvious that $\chi_{56} + \frac{1}{2}i$ commutes with $m_{\mu\nu}$, and it can be proved to commute with $m_{\mu 5}$ as follows. We have

$$\begin{aligned} m_{\mu 5}(\chi_{56} + \tfrac{1}{2}i) &= C(\chi_{\mu 5} - i\chi_{\mu 6})(\chi_{56} + \tfrac{1}{2}i) \\ &= C\{\chi_{56}(\chi_{\mu 5} - i\chi_{\mu 6}) - \chi_{\mu 6} - i\chi_{\mu 5} + \tfrac{1}{2}i(\chi_{\mu 5} - i\chi_{\mu 6})\} \\ &= C(\chi_{56} - \tfrac{1}{2}i)(\chi_{\mu 5} - i\chi_{\mu 6}) \\ &= (\chi_{56} + \tfrac{1}{2}i)m_{\mu 5}. \end{aligned}$$

The supplementary conditions thus pick out a set of ψ's that is invariant under all the operations m_{ab}.

For this set of ψ's, (41) reduces to

$$[m_{\mu 5}, m_{\nu 5}] = m_{\mu\nu},$$

so the m_{ab} satisfy all the commutation relations (2), (3). They thus provide a representation of the 3 + 2 de Sitter group. The representation is unitary, because the m_{ab} are all equal to minus their adjoints, as follows in the case of $m_{\mu 5}$ from C and Ci being self-adjoint.

Some properties of the representation will be worked out. We have, from (30) and (42),

$$m_{12}m_{34} + m_{23}m_{14} + m_{31}m_{24} = 0. \quad (44)$$

The corresponding relations involving other sets of four of the suffixes $1, \cdots 5$ also hold.

The condition (43) means that $i\chi_{56}$ has just the one eigenvalue $\frac{1}{2}$. It follows from the parity discussion in connection with the form (20) that i times any cyclic χ_{AB} has half-odd integral eigenvalues. Thus im_{12}, im_{23}, im_{31} have half-odd integral eigenvalues. It follows then that

$$m_{12}^2 + m_{23}^2 + m_{31}^2 = -k(k + 1), \quad (45)$$

where k has half-odd integral eigenvalues.

We have, from (39) and (14),

$$\begin{aligned} m_{45}^2 &= (\chi_{45} + i\chi_{46})(\chi_{45} - i\chi_{46}) \\ &= \chi_{45}^2 + \chi_{46}^2 - i\chi_{56} \\ &= \chi_{12}^2 + \chi_{23}^2 + \chi_{31}^2 - \chi_{56}^2 - i\chi_{56}, \end{aligned} \quad (46)$$

with the help of (26). Thus, with the supplementary condition (43),

$$m_{45}^2 = m_{12}^2 + m_{23}^2 + m_{31}^2 - \tfrac{1}{4}, \quad (47)$$

$$= -(k + \tfrac{1}{2})^2, \quad (48)$$

with the help of (45). It follows that im_{45} has integral eigenvalues. Thus the representation gives a mixing of integral and half-odd integral eigenvalues for the cyclic rotations.

The k introduced by Eq. (45) may be restricted to have only positive eigenvalues. It follows from (48) that m_{45} does not have the eigenvalue zero.

The absence of the zero eigenvalue has the consequence that the representation can be reduced to two component representations, involving the eigenfunctions of im_{45} with positive and negative eigenvalues, respectively. To prove the result, we note that m_{12}, m_{23}, or m_{31} applied to an eigenfunction of im_{45} does not change the eigenvalue, while $m_{r4}(r = 1, 2, 3)$ applied to it changes the eigenvalue by ± 1, on account of the commutation relation

$$[[m_{r4}, m_{45}], m_{45}] = -m_{r4}. \quad (49)$$

Thus, any application of operators $m_{\mu\nu}$ to an eigenfunction with a positive eigenvalue produces a linear combination of eigenfunctions with positive eigenvalues, as they cannot make the jump from 1 to -1. One can easily check that the two components are irreducible.

The eigenfunctions of im_{45} may be obtained in the following way: We have, from (46) and (25),

$$\begin{aligned} m_{45}^2\psi &= \{-\chi_{56}^2 - i\chi_{56} - \chi_0(\chi_0 + 1) + \tfrac{1}{4}u^\dagger u\partial^\dagger\partial\}\psi \\ &= \tfrac{1}{4}(-1 + u^\dagger u\partial^\dagger\partial)\psi. \end{aligned} \quad (50)$$

Now suppose ψ is of the form

$$\psi = (u^\dagger u)^{-\frac{1}{2}n-1}\phi, \quad (51)$$

where ϕ is homogeneous of degree n. This ψ satisfies (42). Suppose further that ϕ is harmonic, i.e.,

$$\partial^\dagger\partial\phi = 0.$$

Equation (50) now leads to

$$m_{45}^2\psi = -\tfrac{1}{4}(n + 1)^2\psi. \quad (52)$$

The condition (43) leads to

$$(\chi_{56} + \tfrac{1}{2}i)\phi = 0. \quad (53)$$

Thus any harmonic ϕ satisfying (53) provides, according to (51), a ψ that is a linear combination of the eigenfunctions of im_{45} with eigenvalues $\pm\frac{1}{2}(n + 1)$. Of course n has to be an odd integer.

One can infer this directly by noting that (51) is of the form (20) with ϕ for P and the condition (43) requires the function ϕ to be of odd parity.

INHOMOGENEOUS FORM OF THE REPRESENTATION

The representation given by (39) involves the four u's in a homogeneous manner, each $m_{\mu\nu}$ being of degree zero in the u's. One may use the two supplementary conditions (42), (43) to eliminate two of the variables in the wavefunctions, so that there are only two left, which then appear in a nonhomogeneous manner. The resulting form of the representation is more convenient for problems requiring detailed calculation.

Let us first pass from the four real u's to two complex variables. Put

$$v_1 = u_1 + iu_2, \qquad v_2 = u_3 + iu_4.$$

The conjugate complex equations are

$$v_1^* = u_1 - iu_2, \qquad v_2^* = u_3 - iu_4.$$

The operators $\partial/\partial v_1$, $\partial/\partial v_2$ will be written d_1, d_2 for brevity, and they have the values

$$d_1 = \tfrac{1}{2}(\partial_1 - i\partial_2), \qquad d_2 = \tfrac{1}{2}(\partial_3 - i\partial_4).$$

Their conjugates are

$$d_1^* = \tfrac{1}{2}(\partial_1 + i\partial_2), \qquad d_2^* = \tfrac{1}{2}(\partial_3 + i\partial_4).$$

We may choose the γ matrices so that, with the χ's given by (12),

$$\begin{aligned}
\chi_{14} - i\chi_{23} &= -v_2 d_1 - v_1 d_2,\\
\chi_{15} - i\chi_{16} &= v_1^* d_1 - v_2^* d_2,\\
\chi_{24} - i\chi_{31} &= -i(v_2 d_1 - v_1 d_2),\\
\chi_{25} - i\chi_{26} &= i(v_1^* d_1 + v_2^* d_2),\\
\chi_{34} - i\chi_{12} &= v_1 d_1 - v_2 d_2,\\
\chi_{35} - i\chi_{36} &= v_2^* d_1 + v_1^* d_2,\\
\chi_0 + i\chi_{56} &= v_1 d_1 + v_2 d_2,\\
\chi_{45} - i\chi_{46} &= v_2^* d_1 - v_1^* d_2.
\end{aligned} \tag{54}$$

These eight complex equations give sixteen real equations which determine the sixteen γ's, and one finds that the γ's so determined have all the desired properties.

The supplementary conditions (42), (43) may be written

$$\begin{aligned}
(\chi_0 + i\chi_{56})\psi &= -\tfrac{1}{2}\psi,\\
(\chi_0 - i\chi_{56})\psi &= -\tfrac{3}{2}\psi.
\end{aligned} \tag{55}$$

In the new notation they become

$$\begin{aligned}
(v_1 d_1 + v_2 d_2)\psi &= -\tfrac{1}{2}\psi,\\
(v_1^* d_1^* + v_2^* d_2^*)\psi &= -\tfrac{3}{2}\psi.
\end{aligned} \tag{56}$$

They show that ψ is homogeneous of degree $-\frac{1}{2}$ in v_1, v_2 and homogeneous of degree $-\frac{3}{2}$ in v_1^*, v_2^*. Thus ψ is of the form

$$\psi = v_2^{-\frac{1}{2}} v_2^{*-\frac{3}{2}} f(z, z^*), \tag{57}$$

where

$$z = v_1/v_2.$$

Resolving z into its real and pure imaginary parts,

$$z = x_1 + ix_2,$$

we have

$$\psi = v_2^{-\frac{1}{2}} v_2^{*-\frac{3}{2}} g(x_1, x_2). \tag{58}$$

The wavefunction ψ is determined by the function f or g of two variables. Thus we may take f or g to be the wavefunction in a new form of the representation.

Let us see what is the squared length of a wavefunction in the new form. The condition (42) shows that ψ is of the form (18) with $b = 0$, so we have again a squared length like (19) and have to drop out the infinite factor $\int \rho^{-1}\, d\rho$. We may define the variables θ_1, θ_2, θ_3 by means of the equations

$$\begin{aligned}
v_1 &= |v_1| e^{i\theta_1} = \rho \sin\theta_3 e^{i\theta_1},\\
v_2 &= |v_2| e^{i\theta_2} = \rho \cos\theta_3 e^{i\theta_2},
\end{aligned}$$

Thus,

$$|z| = |v_1/v_2| = \tan\theta_3.$$

We now have

$$\begin{aligned}
d^4u &= |v_1|\, d|v_1|\, d\theta_1\, |v_2|\, d|v_2|\, d\theta_2\\
&= |v_1|\, |v_2|\, \rho\, d\rho\, d\theta_3\, d\theta_1\, d\theta_2\\
&= \rho^3\, d\rho \sin\theta_3 \cos\theta_3\, d\theta_1\, d\theta_2\, d\theta_3\\
&= \rho^3\, d\rho \cos^4\theta_3\, d\theta_1\, d\theta_2\, |z|\, d|z|.
\end{aligned}$$

Hence, with ψ of the form (58),

$$\int |\psi|^2\, d^4u = \int \rho^{-1}\, d\rho \int |g|^2\, d\theta_1\, d\theta_2\, |z|\, d\,|z|.$$

The function g involves only $|z|$ and $\theta_1 - \theta_2$, so we may pass from the variables θ_1, θ_2 to $\theta_1 + \theta_2$ and $\theta_1 - \theta_2$, and then carry out the integration with respect to $\theta_1 + \theta_2$. The result is, with omission of the infinite factor $\int \rho^{-1}\, d\rho$,

$$\begin{aligned}
\tfrac{1}{2}\int |g|^2\, d(\theta_1 + \theta_2)\, d(\theta_1 - \theta_2)\, |z|\, d|z|\\
= \pi \int |g|^2\, dx_1\, dx_2.
\end{aligned} \tag{59}$$

Thus we must take the squared length of the new wavefunctions in the usual way, with weight factor unity.

Define the operators

$$D = v_2 d_1, \qquad D^* = v_2^* d_1^*, \qquad C_1 = C v_2^*/v_2. \tag{60}$$

Note that $C_1^2 = 1$,

$$C_1 z C_1 = z^*, \qquad C_1 D C_1 = D^*. \tag{61}$$

With ψ of the form (57), we have

$$\begin{aligned} D\psi &= v_2^{-\frac{1}{2}} v_2^{*-\frac{3}{2}}\, \partial f/\partial z, \\ D^*\psi &= v_2^{-\frac{1}{2}} v_2^{*-\frac{3}{2}}\, \partial f/\partial z^*, \\ C_1\psi &= v_2^{-\frac{1}{2}} v_2^{*-\frac{3}{2}} f^*. \end{aligned} \tag{62}$$

Thus the operators D, D^*, C_1 applied to a ψ of the form (57) give other ψ's of this form, and are equivalent to the operators $\partial/\partial z$, $\partial/\partial z^*$, and taking the conjugate complex, applied to f. So these three operators occur in the expressions for the m_{ab} in the new form of the representation.

One finds from (54), with the help of (56),

$$\begin{aligned} m_{14} - i m_{23} &= (z^2 - 1)D - z(v_1 d_1 + v_2 d_2) \\ &= (z^2 - 1)D + \tfrac{1}{2}z, \\ m_{14} + i m_{23} &= (z^{*2} - 1)D^* - z^*(v_1^* d_1^* + v_2^* d_2^*) \\ &= (z^{*2} - 1)D^* + \tfrac{3}{2}z^*, \\ m_{24} - i m_{31} &= -i(z^2 + 1)D + iz(v_1 d_1 + v_2 d_2) \\ &= -i(z^2 + 1)D - \tfrac{1}{2}iz, \\ m_{24} + i m_{31} &= i(z^{*2} + 1)D^* - iz^*(v_1^* d_1^* + v_2^* d_2^*) \\ &= i(z^{*2} + 1)D^* + \tfrac{3}{2}iz^*, \\ m_{34} - i m_{12} &= 2zD - v_1 d_1 - v_2 d_2 = 2zD + \tfrac{1}{2}, \\ m_{34} + i m_{12} &= 2z^*D^* - v_1^* d_1^* - v_2^* d_2^* \\ &= 2z^*D^* + \tfrac{3}{2}, \\ m_{15} &= C_1\{(z + z^*)D - v_1 d_1 - v_2 d_2\} \\ &= C_1\{(z + z^*)D + \tfrac{1}{2}\}, \\ m_{25} &= -C_1 i\{(z - z^*)D - v_1 d_1 - v_2 d_2\} \\ &= -C_1 i\{(z - z^*)D + \tfrac{1}{2}\}, \\ m_{35} &= C_1\{(1 - zz^*)D + z^*(v_1 d_1 + v_2 d_2)\} \\ &= C_1\{(1 - zz^*)D - \tfrac{1}{2}z^*\}, \\ m_{45} &= C_1\{(1 + zz^*)D - z^*(v_1 d_1 + v_2 d_2)\} \\ &= C_1\{(1 + zz^*)D + \tfrac{1}{2}z^*\}. \end{aligned} \tag{63}$$

These equations give the m_{ab} in the new form. One can easily check that they are equal to minus their adjoints, as they have to be with the form (59) for the squared length.

TRANSFORMATION OF VARIABLES

We shall make a transformation of the operators z, z^*, D, D^* occurring in the expressions (63) for the m_{ab} and get a new representation. Define the operators ζ, ζ^*, Δ, Δ^* according to

$$\zeta^2 = 2D, \qquad \zeta^{*2} = -2D^*, \tag{64}$$

$$\Delta = -z\zeta, \qquad \Delta^* = \zeta^* z^*. \tag{65}$$

ζ and ζ^* commute, and will be taken as the basic variable in terms of which the new wavefunctions are expressed. They are adjoint operators, so they form a pair of conjugate complex variables. We have

$$[\Delta, \zeta^2] = -2[z\zeta, D] = 2\zeta,$$

so

$$[\Delta, \zeta] = 1.$$

Also,

$$[\Delta, \zeta^*] = 0.$$

Thus, Δ may be considered as the operator $\partial/\partial\zeta$. Δ^* is minus the adjoint of Δ, so it may be considered as $\partial/\partial\zeta^*$.

From (61),

$$\zeta^{*2} = -C_1\zeta^2 C_1 = -(C_1\zeta C_1)^2.$$

We may assume

$$\zeta^* = iC_1\zeta C_1, \tag{66}$$

this being consistent with the other equations. Equations (65) now give

$$\begin{aligned} \Delta^* &= iC_1\zeta z C_1 \\ &= -iC_1\zeta\Delta\zeta^{-1}C_1. \end{aligned}$$

Since Δ^* commutes with ζ, we get

$$\begin{aligned} \Delta^* &= -i\zeta^{-1}C_1\zeta\Delta\zeta^{-1}C_1\zeta \\ &= -iC_2\Delta C_2, \end{aligned} \tag{67}$$

where

$$C_2 = \zeta^{-1}C_1\zeta. \tag{68}$$

C_2 has the properties

$$C_2^2 = 1, \qquad C_2 i = -iC_2. \tag{69}$$

From (67),

$$C_2\Delta C_2 = i\Delta^*, \qquad C_2\Delta^* C_2 = i\Delta. \tag{70}$$

From (66),

$$C_2\zeta C_2 = -i\zeta^*, \qquad C_2\zeta^* C_2 = -i\zeta. \tag{71}$$

From (69), (70), (71), it follows that C_2 commutes with each of the following:

$$i\Delta\Delta^*, \quad i\zeta\zeta^*, \quad i(\zeta^*\Delta - \zeta\Delta^*), \quad \zeta^*\Delta + \zeta\Delta^*. \tag{72}$$

From (66) again,

$$C_1\zeta^* = -i\zeta C_1 = \zeta C_1 i,$$

or

$$\zeta^{-1}C_1 = C_1 i\zeta^{*-1}.$$

Thus an alternative form for (68) is

$$C_2 = C_1 i\zeta\zeta^{*-1}. \tag{73}$$

In terms of the new operators ζ, ζ^*, Δ, Δ^*, C_2, Eqs. (63) become

$$\begin{aligned} m_{14} - im_{23} &= \tfrac{1}{2}\{(\Delta\zeta^{-1})^2 - 1\}\zeta^2 - \tfrac{1}{2}\Delta\zeta^{-1} \\ &= \tfrac{1}{2}(\Delta^2 - \zeta^2), \\ m_{24} - im_{31} &= -\tfrac{1}{2}i(\Delta^2 + \zeta^2), \\ m_{34} - im_{12} &= -\zeta\Delta - \tfrac{1}{2}, \end{aligned}$$

and the three adjoint equations. Also,

$$\begin{aligned} m_{15} &= \tfrac{1}{2}C_1\{(-\Delta\zeta^{-1} + \zeta^{*-1}\Delta^*)\zeta^2 + 1\} \\ &= \tfrac{1}{2}C_2 i(\zeta^*\Delta - \zeta\Delta^*), \end{aligned}$$

with the help of (73), and similarly,

$$\begin{aligned} m_{25} &= \tfrac{1}{2}C_2(\zeta^*\Delta + \zeta\Delta^*), \\ m_{35} &= -\tfrac{1}{2}C_2 i(\Delta\Delta^* + \zeta\zeta^*), \\ m_{45} &= \tfrac{1}{2}C_2 i(\Delta\Delta^* - \zeta\zeta^*). \end{aligned}$$

Since C_2 commutes with the expressions (72), it commutes with the $m_{\mu 5}$. It must therefore also commute with the $m_{\mu\nu}$, which are just the commutators of the $m_{\mu 5}$. Let us define new infinitesimal operators m'_{ab} by

$$\begin{aligned} m'_{\mu\nu} &= m_{\mu\nu}, \\ m'_{\mu 5} &= C_2 m_{\mu 5}. \end{aligned}$$

One sees at once that the m'_{ab} also satisfy the commutation relations (2), (3) and thus provide a representation of the 3 + 2 de Sitter group. The C_2 operator does not occur in the m'_{ab}, each of them being merely a quadratic function of the variables ζ, ζ^*, Δ, Δ^*.

The new representation can be better understood if we refer it to real variables instead of the complex ζ, ζ^*. Put

$$\zeta = 2^{-\frac{1}{2}}(q_1 + iq_2), \qquad \zeta^* = 2^{-\frac{1}{2}}(q_1 - iq_2),$$

$$-i\Delta = 2^{-\frac{1}{2}}(p_1 - ip_2), \qquad -i\Delta^* = 2^{-\frac{1}{2}}(p_1 + ip_2).$$

Thus q_1, q_2, p_1, p_2 are Hermitian operators satisfying

$$[q_1, p_1] = [q_2, p_2] = i,$$

with their other commutators vanishing, so they are like canonical variables in quantum mechanics. We now get

$$\begin{aligned} im'_{12} &= \tfrac{1}{2}(q_1p_2 - q_2p_1), \\ im'_{23} &= \tfrac{1}{4}(p_1^2 + q_1^2 - p_2^2 - q_2^2), \\ im'_{31} &= -\tfrac{1}{2}(q_1q_2 + p_1p_2), \\ im'_{14} &= \tfrac{1}{2}(q_1q_2 - p_1p_2), \\ im'_{24} &= \tfrac{1}{4}(q_1^2 - p_1^2 - q_2^2 + p_2^2), \\ im'_{34} &= \tfrac{1}{2}(q_1p_1 + p_2q_2), \\ im'_{15} &= -\tfrac{1}{2}(q_1p_2 + q_2p_1), \\ im'_{25} &= -\tfrac{1}{2}(q_1p_1 - q_2p_2), \\ im'_{35} &= \tfrac{1}{4}(q_1^2 + q_2^2 - p_1^2 - p_2^2), \\ im'_{45} &= \tfrac{1}{4}(p_1^2 + q_1^2 + p_2^2 + q_2^2). \end{aligned}$$

There are ten independent quadratic functions of the four variables q_1, q_2, p_1, p_2, and suitable linear combinations of these ten provide the expressions for the m'_{ab}.

The algebraic connection between the operators in the new representation and those of the previous representation (39) or (63) ensures that the new operators satisfy the same algebraic relations as before, e.g., the commutation relations (2), (3) and the various quadratic relations such as (47). However, the new representation is not equivalent to the previous one.

The new representation can be reduced to two component representations, for which the wavefunctions $\psi(q_1, q_2)$ are even or odd functions of the q's, respectively. One can easily check that these components are irreducible.

One sees that im'_{12} has integral or half-odd integral eigenvalues according to whether ψ is even or odd. Further, the expression for im'_{45} is like half the energy of two harmonic oscillators. Its eigenvalues are all positive and the lowest eigenvalue is $\frac{1}{2}$, coming from the zero-point energy of each oscillator. Even wavefunctions lead to half-odd integral eigenvalues for im'_{45}, and odd wavefunctions lead to integral eigenvalues. Thus for each of the component representations, one of the operators im'_{12}, im'_{45} has integral eigenvalues, and the other half-odd integral eigenvalues. The component with im'_{12} half-odd and

im'_{45} integral is equivalent to the component of (39) or (63) for which im_{45} has positive eigenvalues.

The new representation is thus more fundamental than the previous one in two respects—it separates the negative eigenvalues of im_{45} away from the positive ones and it allows both the alternatives of integral and half-odd eigenvalues for im_{45}. The greater power of the new representation comes from the introduction, by (64), of new variables equal to the square roots of previous variables.

The quadratic functions of the four variables q_1, q_2, p_1, p_2 are the infinitesimal generators of the group of linear transformations of these variables that leave their commutation relations invariant. It has been pointed out to me by R. Jost that this group is just the 4-dimensional simplectic group, which is equivalent to the 3 + 2 de Sitter group. These results are easily established by the following argument:

For convenience, put $p_1 = q_3$, $p_2 = q_4$. We then have to consider linear transformations of the four q's which leave the commutators $[q_m, q_n]$ invariant ($m, n = 1, 2, 3, 4$). Let a_m be a set of four numbers which transform contravariantly to the four q's, so that $a_m q_m$ is invariant, and let b_m be another such set of four numbers. Put

$$W_{mn} = a_m b_n - a_n b_m.$$

Thus, each linear transformation of the q's gives rise to a linear transformation of the a's and b's, which in turn gives rise to a linear transformation of the W's.

We have

$$[a_m q_m, b_n q_n] = a_m b_n [q_m, q_n] = 2i(W_{13} + W_{24}).$$

This is invariant. Also,

$$W_{12}W_{34} + W_{23}W_{14} + W_{31}W_{24} = 0.$$

Put

$$W_{12} = x_1 + x_4, \quad W_{23} = x_2 + x_5, \quad W_{31} = x_3 + x_6,$$

$$W_{34} = x_1 - x_4, \quad W_{14} = x_2 - x_5, \quad W_{24} = x_3 - x_6.$$

Then the six x's are subject to linear transformations which leave x_6 invariant, and which preserve the quadratic equation

$$x_1^2 + x_2^2 + x_3^2 - x_4^2 - x_5^2 - x_6^2 = 0.$$

This is just the 3 + 2 de Sitter group.